

Immunoinformatics-driven design and construction of a multi-epitope peptide vaccine candidate against rabies virus

ABSTRACT

Rabies, a lethal zoonotic disease caused by lyssaviruses, poses a global threat, with approximately 60,000 deaths annually, predominantly affecting rural populations in Asia and Africa. Despite available vaccines, the disease remains underreported, with up to 99% of deaths attributed to the classical rabies virus transmitted by dogs. Clinical symptoms usually lead to fatality, and no reliable treatment exists post-onset. This study aimed to design a novel multi-epitope vaccine against rabies using immunoinformatic analyses targeting glycoprotein (G), matrix protein (M), nucleoprotein (N), and phosphoprotein (P). The IEDB server predicted B and T cell epitopes and assessed for antigenicity, allergenicity, toxicity, and stability using VaxiJen, Allertop, ToxinPred, and ProtParam. Selected epitopes were linked, conjugated with an adjuvant, and evaluated for global population coverage (99.74%) using IEDB. Molecular modeling via Robetta, docking via HADDOCK, and dynamics simulations via GROMACS ensured construct stability. This study introduces a promising vaccine candidate warranting further pre-clinical evaluation for prophylactic use.

INTRODUCTION

- The disease transmits through the saliva of infected animals (mostly dogs), and travels from the bite wound to the nervous system, causing a delayed onset of flu-like symptoms that progress to fatal neurological damage, including anxiety, hydrophobia, phonophobia, and paralysis. Early detection and post-exposure treatment are crucial, as there's no effective treatment once symptoms appear.
- Rabies, a zoonotic disease primarily transmitted by dogs, remains a devastating threat. Over 60,000 people, especially in poor rural communities, with children under the age of 15 accounting for about half of cases, lose their lives to this viral encephalitis each year.
- Despite current vaccinations that are concentrated, pure cell culture and embryonated egg-based rabies vaccines (CCEVs) have been shown to be exceptionally safe and efficacious in preventing rabies. However, further research on more effective and accessible vaccines like multi-epitope peptide vaccines, potentially offering broader protection and long-lasting immunity, is necessary.

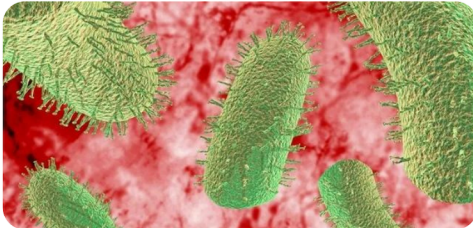
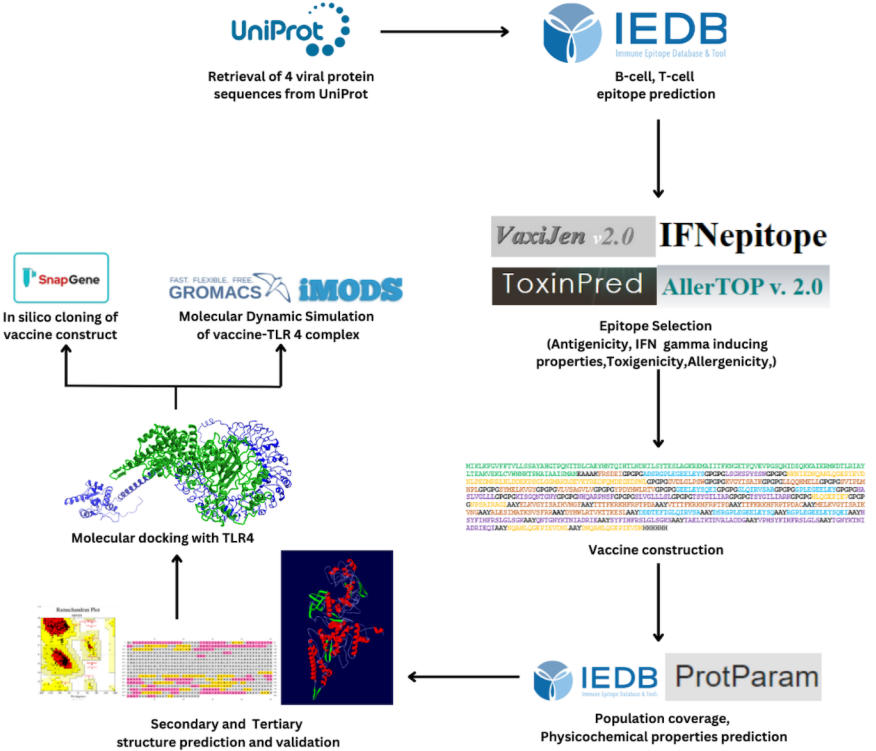


Fig 1: Rabies virus

MATERIALS AND METHODS

An integrated in silico method was utilized in the construction of the vaccine model.



RESULTS

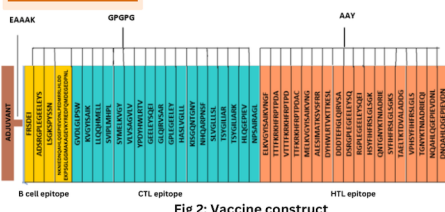


Fig 2: Vaccine construct

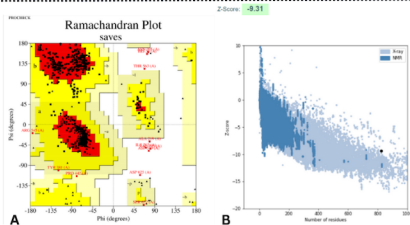


Fig 6 : Validation studies of vaccine construct.

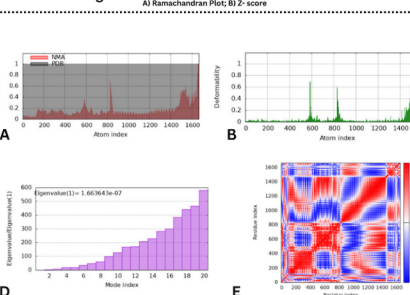


Fig 9 : Results of molecular dynamic simulation study of vaccine - TLR4 docked complex, A) Ramachandran Plot; B) Deformation plot; C) Variance plot; D) Eigenvalue; E) Covariance matrix analysis; F) Elastic network model

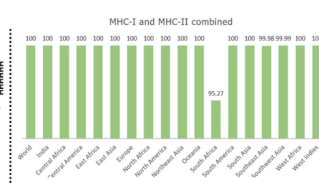


Fig 3 : Population coverage

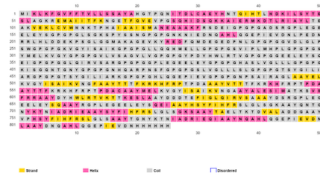


Fig 4 : Secondary structure of vaccine model

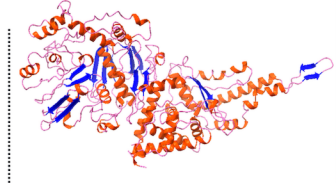


Fig 5 : Tertiary structure of vaccine model

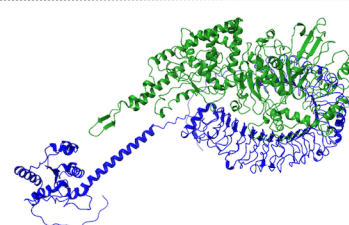


Fig 7 : Interaction between vaccine candidate and TLR4

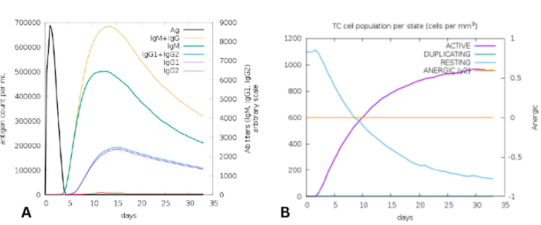


Fig 8 : Immune simulation prediction

A) Antibody response; B) Cytotoxic T cell population per state

CONCLUSION

- Even though modern cell culture and embryonated egg-based rabies vaccines offer safe, effective, and long-lasting protection, research on next-generation multi-epitope vaccines holds promise for even broader and potentially longer-lasting immunity.
- This study designed a novel rabies vaccine construct targeting four key viral proteins: glycoprotein, matrix protein, nucleoprotein, and phosphoprotein.
- In silico analysis suggests promising immunogenicity, requiring further in vitro and in vivo studies to validate the results.

Fig 10 : In silico cloning of RBV vaccine sequence in pET-28a(+) plasmid



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Certificate of Participation

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